
Quantum Information

Exercise 3

1. Mutually unbiased bases

Show that the X , Y and Z bases are all unbiased with respect to each other: each state for one basis results in a completely random result for the other two bases.

2. Shifting certainty

The state of a single qubit is characterized by three numbers, $\langle \sigma^x \rangle$, $\langle \sigma^y \rangle$ and $\langle \sigma^z \rangle$, defined as

$$\langle \sigma^\alpha \rangle = p_0^\alpha - p_1^\alpha$$

Calculate these for the following states and verify that $\langle \sigma^x \rangle^2 + \langle \sigma^y \rangle^2 + \langle \sigma^z \rangle^2$ remains constant.

a)

$$|\psi\rangle = \cos(\theta) |0\rangle + \sin \theta |1\rangle$$

b)

$$|\psi\rangle = \frac{|0\rangle + e^{i\theta} |1\rangle}{\sqrt{2}}$$

2. A useful matrix

Find a matrix σ^z such that

$$\langle \sigma^z \rangle = \langle \psi | \sigma^z | \psi \rangle \quad \forall |\psi\rangle$$